Contextuality and Exact Information Resources over Finite Chain Rings

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Provisional article skeleton: theorem provenance retained

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Abstract

We study possible-outcome support for bipartite matrices over finite commutative chain rings, using explicitly specified complete local basis measurements. The proposed article centers on a classification by the number of nonzero Smith factors and the multiplicity of the least valuation. Under a matched invertible-orbit communication task, the latter invariant gives an exact dh-message codebook bound. Universal exact teleportation instead requires invertibility and an admitted complete analyzer, giving sharp resource separations. We compare shared-ring and literal field tensors, explain the collapse of restricted distinctions under enlarged operations, and give a chosen symbolic-support interface back to Boolean logical matrices. These results concern algebraic support, not Born probabilities or a closed unrestricted process theory. The final introduction must distinguish candidate new classification/synthesis from standard ring and modal ingredients after the remaining theorem-level priority review.

Drafting instruction. This is a build-ready outline, not a completed submission or a claim of established priority. Canonical proof sources are in `../../math/THEOREM_BANK.tex`; cited section labels below are stable source anchors, not this skeleton's theorem numbers. Follow PAPER_SCOPE.md and the claim ownership matrix.

1 Introduction and the operational question

Drafting instruction. Open with the minimal dual-number resource εI_2: four exact codewords and strong support contextuality, but no universal exact raw-state teleportation. Immediately state that it is nonprimitive. Then present the classification and its operation-dependent meaning. Explain why neither generic modal entanglement nor a circulant-coordinate change is a new claim. Cite Paper B only for the Boolean operator foundation.

The primary contribution paragraph should contain three connected assertions: a complete-basis contextuality classification, the matched dh codebook formula, and a contrast with universal teleportation. Tensor choice, enlarged operations and closure limitations delimit these assertions. The arbitrary-party two-setting theorem identifies a separate measurement-setting boundary.

2 Algebra, effects and task contracts

Let K be a finite commutative chain ring with radical (π), nilpotency length s and residue field k. A nonzero resource matrix has Smith exponents $a_1 \le \cdots \le a_r < s$; zero factors have exponent s. Set $h = |\{j : a_j = a_1\}|$. Define primitive projective rows as unit-scaling classes of rows with a unit coordinate. Settings are complete invertible bases; a supported event has nonzero contraction.

Drafting instruction. Insert precise definitions of global assignment and extension of a seed event. State whether the same ray in different settings is an additional identified measurement: the canonical theorem uses basis settings without adding further identification constraints. State DC and TP separately from `../../math/model_contracts.tex`. State exact raw-vector recovery and reference preservation. Do not imply a probability distribution.

2.1 A concrete rotation-derived restriction

The coefficient rotation gives $\mathcal{A} = \mathbb{F}_2[R] \cong \mathbb{F}_2[u]/(u^4)$, $u = I + R$. This is a standard cyclic/chain algebra; it motivates a specified $\mathrm{GL}_m(\mathcal{A})$ restriction rather than deriving a tensor or measurement rule. The lift is not a homomorphism for the original CM product.

Drafting instruction. Import ROT-001–009 as labeled in the registry. Keep this motivation short; move detailed filtration/Boolean-degree calculations to a supplement.

Figure specification: Typed model diagram Separate Paper B's Boolean formulas, coefficient rotation, chosen scalar algebra, shared versus literal tensor, admitted gates, admitted effects and support rule. Identify the choices explicitly.

Figure 1: Planning placeholder; replace with the generated, source-linked figure.

3 The complete-basis classification

Canonical IDs: MOD-003.

Lemma 3.1 (Residue-hyperplane criterion)**.** *Over a finite commutative local ring, a global assignment exists precisely when there are residue hyperplanes on both sides whose complementary primitive lifts have uniformly nonzero cross contractions.*

Drafting instruction. Insert the full selected-ray/complement-span proof from theorem-bank `sec:context`. Review locality versus chain-ring assumptions separately.

Canonical IDs: MOD-005.

Lemma 3.2 (Uniform Hardy event)**.** *For $D = \mathrm{diag}(\pi^a, \pi^b)$ with $a \le b < s$, write $c = \pi^{b-a}$. The displayed flip, shear and lower shear with entry $-c$ produce the supported Hardy event in the canonical four-context support table.*

Drafting instruction. Print all four matrices and the seed-forcing contradiction; retain $-c$ in odd characteristic. Do not substitute a finite-field illustration for the valuation-uniform statement.

Canonical IDs: MOD-004.

Theorem 3.3 (Smith contextuality trichotomy)**.** *Under all complete primitive projective local bases, a nonzero resource is local if $r = 1$, logical but not strongly contextual if $r \geq 2$ and $h = 1$, and strongly contextual if $h \geq 2$.*

Drafting instruction. Insert all three proof directions and the embedded Hardy argument. The new independent dual-number SCC result is validation, not the proof. Add the complete15-class A table only after the theorem.

Canonical IDs: MOD-007.

Theorem 3.4 (Two binary settings)**.** *Any nonzero multi-party $\mathbb{F}_2$ tensor with at most two complete binary settings per site has a global assignment. The statement extends to the specified local rings with residue $\mathbb{F}_2$.*

Drafting instruction. Use the inclusion-minimal coefficient construction and leading-layer functional extension from `sec:settings`. Say explicitly that two-setting logical contextuality remains possible.

4 Exact communication and leading layers

Canonical IDs: RES-001, RES-002.

Theorem 4.1 (Fixed-orbit codebook)**.** *For $0 \neq M \in \mathrm{Mat}_d(K)$, $d \geq 2$, invertible local encoding followed by one arbitrary complete K-linear joint decoder permits exactly dh distinguishable messages. Advantage over d is equivalent to strong contextuality under the matched contracts.*

Drafting instruction. Insert the residue-span upper bound and invertible-encoder/basis-lift construction from `sec:resources`. Give at least one actual decoder and link all fourteen certificates. Call this a message maximum for a task, not an unqualified channel capacity or entropy.

Canonical IDs: RES-003, RES-004, RES-007.

Proposition 4.2 (Teleportation boundary)**.** *Every nonzero universally correct raw-state branch has invertible transfer $T = M^T E^T$; in particular M is invertible. Conversely an invertible resource and an admitted complete basis of invertible reshaped effects give exact correction. The supplied complete analyzers cover dimensions two and eight.*

Drafting instruction. Prove collision in a singular branch, completeness, inverse correction and reference identity. Include $\pi^a I_d$ for $0 < a < s$, mark it nonprimitive, and contrast a primitive dimension-three example. Do not silently claim unrestricted analyzer admissibility in every dimension.

Figure specification: Resource hierarchy
Show r, h, primitive/nonprimitive layers, dh, invertibility and the excluded zero resource. Include the two separators.

Figure 2: Planning placeholder; replace with the generated, source-linked figure.

5 Changing the model changes the resource

Canonical IDs: MOD-001, MOD-002, MOD-009, MOD-010; RES-005, RES-006, RES-008.

Drafting instruction. Prove the balanced-tensor zero-divisor witness and conditioning counterexample. Explain regular binary rank and full-group collapse. Include $\mathrm{diag}(1, 1, u^3)$ and $\mathrm{diag}(1, u, u^2)$: both rank nine after regular expansion, but different h. Explain the specific restricted bipartite table isomorphism by Bob-side transpose involution; it does not contradict tensor inequivalence. Keep field extraction as a rectangular-filter identity, not a probability or a complete distillation protocol.

6 A chosen symbolic-support interface

Canonical IDs: LMB-001–008.

Proposition 6.1 (Enriched support)**.** *After choosing $\mathcal{A}$, valuations extend to $\mathcal{B} \otimes_{\mathbb{F}_2} \mathcal{A} \cong \mathcal{B}[u]/(u^4)$. For $q = \sum_j f_j u^j$, the formula $\bigvee_j f_j$ evaluates true exactly when the valued ring coefficient is nonzero. This applies to compatible enriched contractions.*

Drafting instruction. Import Paper B's pairing naturality rather than re-proving it as new. State the local-ring idempotent obstruction and all additional choices. Put $B^{-1}P_iB$ update identities in an appendix. The interface is not an unrestricted closed process theory or a claim about mixed-characteristic F2-algebra enrichment.

7 Evidence, related work and limitations

Compare MQT and generalized modal measurements [3, 4], the established contextuality hierarchy [11], finite-chain-ring module work [12], ring computation [5], and phase/process frameworks [16, 17, 18, 31].

Drafting instruction. Replace this navigation paragraph with theorem-level comparisons. Present candidate priority as unresolved until checked. Distinguish the102 pytest tests, source-script enumerations, and newly authored SCC/decoder checks. State all operation/measurement/primitivity restrictions in the main limitations section. Exclude Born probabilities and physical realizability.

A Proof and certificate supplement

Drafting instruction. Include basis counts, explicit decoder/analyzer matrices, field rank extraction, local-ring two-setting details, symbolic update identities and a claim-ID-to-final-theorem crosswalk. Preserve original sources and fresh rerun manifests. Do not repeat P02 query proofs.

References

[1] Brian Theory. *Operator-Level Boolean Computation with Correspondence Matrices*. 2026. Unpublished manuscript supplied with this consolidation.

[2] Brian Droncheff. *Correspondence Matrices; Algorithms for Propositional Logic*. 2018. https://doi.org/10.13140/RG.2.2.28036.37764.

[3] Benjamin Schumacher and Michael D. Westmoreland. *Modal quantum theory*. 2010. https://arxiv.org/abs/1010.2929.

[4] Benjamin Schumacher and Michael D. Westmoreland. *Almost quantum theory*. 2012. https://arxiv.org/abs/1204.0701.

[5] Niel de Beaudrap. *On computation with 'probabilities' modulo k*. 2014. https://arxiv.org/abs/1405.7381.

[6] Jeremiah Willcock and Amr Sabry. *Solving UNIQUE-SAT in a Modal Quantum Theory*. 2011. https://arxiv.org/abs/1102.3587.

[7] Roshan P. James and Gerardo Ortiz and Amr Sabry. *Quantum Computing over Finite Fields*. 2011. https://arxiv.org/abs/1101.3764.

[8] Andrew J. Hanson and Gerardo Ortiz and Amr Sabry and Jeremiah Willcock. *The Power of Discrete Quantum Theories*. 2011. https://arxiv.org/abs/1104.1630.

[9] Robert Beals and Harry Buhrman and Richard Cleve and Michele Mosca and Ronald de Wolf. *Quantum Lower Bounds by Polynomials*. 2001. https://arxiv.org/abs/quant-ph/9802049.

[10] Phillip Diamond and Benjamin Schumacher. *Cloning, deleting, and hiding in modal quantum theory*. 2023. https://arxiv.org/abs/2310.04397.

[11] Samson Abramsky and Adam Brandenburger. *The Sheaf-Theoretic Structure of Non-Locality and Contextuality*. 2011. https://arxiv.org/abs/1102.0264.

[12] Eimear Byrne and Anna-Lena Horlemann and Karan Khathuria and Violetta Weger. *Density of Free Modules over Finite Chain Rings*. 2021. https://arxiv.org/abs/2106.09403.

[13] Heinz Lueneburg. *On a decomposition of square matrices over a ring with identity*. 1988. https://mat.univie.ac.at/~slc/opapers/sc18luene.pdf.

[14] Harinaivo Andriatahiny. *The Generalized Reed-Muller codes and the radical powers of a modular algebra*. 2016. https://arxiv.org/abs/1601.07633.

[15] Henry Chimal-Dzul and Niklas Gassner and Joachim Rosenthal and Reto Schnyder. *Efficient Description of some Classes of Codes using Group Algebras*. 2022. https://arxiv.org/abs/2208.04061.

[16] Bob Coecke and Bill Edwards and Robert W. Spekkens. *Phase groups and the origin of non-locality for qubits*. 2010. https://arxiv.org/abs/1003.5005.

[17] Stefano Gogioso. *Fantastic Quantum Theories and Where to Find Them*. 2017. https://arxiv.org/abs/1703.10576.

[18] Kenta Cho and Bart Jacobs and Bas Westerbaan and Abraham Westerbaan. *An Introduction to Effectus Theory*. 2015. https://arxiv.org/abs/1512.05813.

[19] Samson Abramsky and Bob Coecke. *A categorical semantics of quantum protocols*. 2004. https://arxiv.org/abs/quant-ph/0402130.

[20] Eric Chitambar and Felix Leditzky. *On the Duality of Teleportation and Dense Coding*. 2023. https://arxiv.org/abs/2302.14798.

[21] Daniel Gottesman. *Stabilizer Codes and Quantum Error Correction*. 1997. https://arxiv.org/abs/quant-ph/9705052.

[22] Charles H. Bennett. *Logical Reversibility of Computation*. 1973. https://www.cs.princeton.edu/courses/archive/fall04/cos576/papers/bennett73.html.

[23] Alan Mishchenko and Satrajit Chatterjee and Robert K. Brayton. *DAG-aware AIG rewriting: A fresh look at combinational logic synthesis*. 2006. https://people.eecs.berkeley.edu/~alanmi/publications/2006/dac06_rwr.pdf.

[24] Daizhan Cheng and Yin Zhao and Xiangru Xu. *Matrix Approach to Boolean Calculus*. 2011. https://lsc.amss.ac.cn/~dcheng/preprint/DerBool_cdc11.pdf.

[25] Eduardo Mizraji. *Vector Logic: A Natural Algebraic Representation of the Fundamental Logical Gates*. 2008. https://doi.org/10.1093/logcom/exm057.

[26] Zeno Toffano. *Eigenlogic in the spirit of George Boole*. 2015. https://arxiv.org/abs/1512.06632.

[27] Michael Brickenstein and Alexander Dreyer. *PolyBoRi: A framework for Groebner-basis computations with Boolean polynomials*. 2009. https://polybori.sourceforge.net/doc.html.

[28] C. R. Edwards. *The logic of Boolean matrices*. 1972. https://doi.org/10.1093/comjnl/15.3.247.

[29] August Stern. *Matrix Logic: Theory and Applications*. 1988. https://doi.org/10.1016/C2009-0-14445-8.

[30] Randal E. Bryant. *Graph-based algorithms for Boolean function manipulation*. 1986. https://doi.org/10.1109/TC.1986.1676819.

[31] J. R. B. Cockett and Stephen Lack. *Restriction categories I: categories of partial maps*. 2002. https://cspages.ucalgary.ca/~robin/FMCS/FMCS_06/RestrictionsI.pdf.